

Computational Analysis of Multimodal Virtual Reality Behavioral Data

1: Executive Summary

This proposal outlines a computational analysis of the multimodal game data from the UCSD Brain Health VR Study (PI: Dr. Judy Pa), comparing the two VR arms – VR-seated and VR + cycling – across 80 participants, and 48 longitudinal visits each. The dataset combines continuous 18 Hz multimodal recordings (3D position, head pose, heart rate, bicycle telemetry) with millisecond-resolution Go/No-Go decisions in a foraging task embedded in a branching maze. The combined volume across the cognitive arms exceeds 19,000 trials, 830 million continuous samples, and 3.8 million decision events.

The primary aim is unsupervised navigational phenotyping: recovering interpretable latent strategies from trajectory and head-pose data via variational autoencoders, density-based clustering, and Hidden Markov state segmentation, with longitudinal phenotype dynamics tested across intervention arms. Secondary aims layer hierarchical drift-diffusion modeling of decision data, reinforcement-learning analysis of navigation choices, and generalized-additive-model tests of arousal-cognition coupling, all within a hierarchical Bayesian inferential framework. The full pipeline is delivered as a containerized open-source monorepo with continuous integration, experiment tracking, and an interactive dashboard.

2: Background/Motivation

Acute aerobic exercise reliably enhances executive function and hippocampal-dependent memory (Hillman, Erickson, & Kramer, 2008; Voss et al., 2013), and immersive virtual reality engages the same hippocampal-entorhinal network implicated in rodent watermaze paradigms (Ekstrom et al., 2003; Jacobs et al., 2013). The intersection, exercise performed during VR navigation, represents both a methodological frontier and a translational opportunity. Existing combined-modality studies are limited by small samples, single-session designs, and outcome measures restricted to total accuracy or completion time. The latent cognitive processes that generate those outcomes – strategy preference, learning trajectory, decision dynamics, arousal–cognition coupling – remain largely uncharacterized.

The present dataset addresses each of these limitations directly: 80 participants across two VR intervention arms, 48 longitudinal visits per participant, and continuous multimodal recordings paired with discrete decision data. The analysis exploits this density to apply unsupervised representation learning, mechanistic cognitive models, and hierarchical Bayesian inference within a single integrated pipeline, supporting both behavioral discovery and adequately powered statistical inference.

3: Goals/Hypotheses

The project is organized around four aims with explicit priority ordering. The first two are primary; the remaining two are contingent on phenotyping completion.

Aim 1 (Primary): Unsupervised navigational phenotyping

Unsupervised phenotyping will identify latent navigation strategies, which may correspond to landmark-guided, route-based, or mixed navigation styles. and head-pose data using variational-autoencoder representations, HDBSCAN clustering, and Hidden Markov state segmentation. Validated via bootstrap adjusted Rand index (target > 0.6) and reproducibility on a 20% subject holdout sequestered at project initiation.

Aim 2 (Primary): Longitudinal phenotype dynamics

Test whether phenotype membership shifts across the 48 visits and whether the shift trajectories differ between intervention arms. Tested via multinomial mixed-effects regression with arm $\times$ visit interaction and Markov-style phenotype-transition modeling.

Aim 3 (Secondary): Cognitive process modeling

Apply the hierarchical Drift-Diffusion Model (HSSM) to Go/No-Go reaction-time data to dissociate evidence-accumulation capacity from response caution. In parallel, fit Q-learning and a Daw-style model-based/model-free hybrid (Daw, Niv, & Dayan, 2005) to navigation choices at maze forks; the mixing parameter ω is a known biomarker of cognitive flexibility modulated by arousal (Otto et al., 2013).

Aim 4 (Secondary): Arousal-cognition coupling

Within the VR + cycling arm, test whether within-trial heart rate modulates cognitive performance with the inverted-U signature predicted by Yerkes-Dodson. Generalized additive models with breakpoint regression as a complementary specification.

All inferential analyses are conducted within a hierarchical Bayesian framework (PyMC/Bambi) with random subject intercepts and slopes accommodating the nested trial-in-visit-in-subject-in-arm structure. Confirmatory analyses are pre-specified before fitting and run on the held-out subject sample.

4: Dataset Description

The dataset consists of game data from 80 participants across two VR arms: VR-seated ($n = 40$) and VR + cycling ($n = 43$), each completing 48 longitudinal visits with approximately five trials per visit and 25 to 45 minutes per visit. Each trial generates continuous 18 Hz multimodal recordings: 3D position and its derivatives, head pose along three Euler axes, heart rate (cycling arm only), and bicycle telemetry (cadence, power, torque, calorie rate, resistance; cycling arm only). The exercise-only arm of the parent study generates no game data and is outside the present scope. An accompanying event-level file records each object encountered during a trial: category, target indicator, entry and exit timestamps, decision time in milliseconds, and spatial side of stimulus and response. The combined volume is approximately 19,200 trials, 830 million continuous samples, 3.8 million decision events, and 88 GB of raw CSV, reducing to roughly 10 GB in compressed Parquet.

5: Analytical Approach

Data engineering

Raw CSVs are inventoried into a manifest table keyed on the (subject, visit, trial) tuple and validated against schemas defined in Pandera. Sampling-rate jitter is corrected by uniform resampling to 20 Hz; head-rotation channels are unwrapped to remove the 0° to 360° discontinuity; out-of-range heart-rate samples are flagged. Continuous trajectories are snapped to a manually annotated canonical maze graph (NetworkX) and reparameterized as (edge identifier, fractional position), replacing continuous coordinates with the symbolic state-space the maze actually instantiates. Processed data is partitioned into Parquet files indexed under DuckDB for analytical querying.

Phenotype discovery (Aim 1)

Three complementary unsupervised methods are layered. A variational autoencoder compresses whole-trial trajectories – graph-snapped position, head yaw, velocity – into a low-dimensional latent representation. HDBSCAN clusters the latent vectors to identify trajectory phenotypes without pre-specifying cluster count. Hidden Markov Models segment within-trial behavior into latent states (exploration, decision-making, exploitation, disengagement), providing a complementary segmentation that informs phenotype interpretation. Cluster validity is assessed via bootstrap adjusted Rand index (100 resamples), silhouette score, and reproducibility across a stratified 80/20 train/test split. Hyperparameter selection: latent dimensionality, HDBSCAN min-cluster-size, HMM state count, uses cross-validated reconstruction loss and stability metrics.

Cognitive process modeling (Aim 3)

The Drift-Diffusion Model is fitted via HSSM (the modern hierarchical-Bayesian successor to HDDM; Wiecki, Sofer, & Frank, 2013), decomposing reaction-time distributions into drift rate v , boundary separation a , non-decision time t_0 , and starting-point bias z . Hierarchical regression structure permits arm, visit, and within-trial heart rate to enter as covariates on each parameter, directly testing whether exercise alters capacity, caution, or both. Sequential navigation

decisions at maze forks are modeled with subject-level Q-learning baselines and a Daw-style hybrid; the model-based/model-free mixing parameter ω yields a per-subject biomarker of cognitive flexibility.

Arousal–cognition coupling (Aim 4)

Within the VR + cycling arm, generalized additive models fit trial-level accuracy and drift rate as smooth functions of within-trial heart rate; breakpoint regression locates the inflection of any inverted-U signature.

Confirmatory inference

All hypothesis tests are conducted in a hierarchical Bayesian framework using PyMC and Bambi, with random subject intercepts and slopes accommodating the nested trial-in-visit-in-subject-in-arm structure. Posterior predictive checks, $\hat{R}$ diagnostics, energy plots, and equivalence tests are reported via ArviZ. Multiple outcomes are modeled jointly through correlated random effects.

Engineering and reproducibility

The pipeline is implemented as a containerized monorepo (Docker), with deterministic dependency management (uv), data and model versioning (DVC), experiment tracking (MLflow), and continuous integration (GitHub Actions enforcing ruff, mypy, pytest, and schema validation on every commit). A Streamlit dashboard supports per-trial trajectory replay, phenotype exploration, and posterior visualization. A single GNU make target reproduces the full analysis from raw data to figures.

6: Deliverables

Manuscript

Preprint reporting the discovery and validation of latent navigation phenotypes from multimodal VR behavioral data, with longitudinal phenotype dynamics across the two intervention arms. Stretch contributions from the cognitive-process and physiological-coupling analyses are integrated into the same manuscript or split into a companion methods note depending on results.

Open-source pipeline

A containerized monorepo released under MIT license with deterministic dependency management, schema specifications, documentation, and a single make target reproducing the full analysis from raw data to figures.

Interactive dashboard

A Streamlit application deployed to a public URL, supporting per-trial trajectory replay, phenotype exploration, and posterior visualization.

Reproducibility artifacts

Pre-registered analysis plan on the Open Science Framework, MLflow experiment logs, DVC-versioned data and models, and continuous-integration test reports.